Lorentzian extension of the four-witness Wick-rotation theorem on truncated $S^3 \times \mathbb{R}$ spectral triples at finite cutoff

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Abstract

The four-witness Wick-rotation theorem (Hartle–Hawking [11], Sewell [21], Bisognano–Wichmann [1], Unruh [25]) was closed at the operator-system level on the Riemannian truncated Camporesi–Higuchi spectral triple over SU(2) in the companion Paper 42 [31]. The present paper extends that closure to signature $(s,t) = (3,1)$ West-coast on the Krein space $\mathcal{K}_{n_{\max},N_t} := \mathcal{H}_{\mathrm{GV}}^{n_{\max}} \otimes \mathbb{C}^{N_t}$ via the van den Dungen 2016 [26] Proposition 4.1 lift and the Bizi–Brouder–Besnard 2018 [2] (m,n) classification at $(4,6)$. On a hemispheric wedge $W_L = P_W^{\mathrm{spatial}} \otimes P_{t \geq 0}$ of $S^3 \times \mathbb{R}$ with the Bisognano–Wichmann choice of local Hamiltonian $H_{\mathrm{local}} := K_L^{\alpha,W}/\beta$, we establish: (i) the geometric BW-α modular Hamiltonian $K_L^\alpha = J_{\mathrm{polar}} \otimes I_{N_t}$ has integer spectrum and produces the bit-exact period closure $\sigma_{2\pi}^{L,\alpha}(O) = O$; (ii) the Tomita–Takesaki BW-γ modular Hamiltonian $K_L^{\mathrm{TT}} = -\log \Delta_L$ obtained from polar decomposition of the modular S operator on the GNS Hilbert–Schmidt space of the wedge KMS state $\rho_W^L = e^{-K_L^{\alpha,W}}/Z$ produces the bit-exact period closure $\sigma_{2\pi}^{L,\mathrm{TT}}(a) = a$; (iii) the flow conjugacy $\sigma_t^{L,\mathrm{TT}}(a) = \sigma_{-t}^{L,\alpha}(a)$ holds bit-exact at general t; (iv) the six-witness collapse on the BW-aligned wedge KMS state is bit-exact, with ρ_W^L itself β-independent. All four results hold at every tested $(n_{\max}, N_t) \in \{1,2,3\} \times \{1,11,21\}$. The main substantive structural finding is that the load-bearing $H_{\mathrm{local}} \neq D_L^W$ scope finding of Paper 42 (the spectral-action Dirac is not the modular-Hamiltonian generator on the Bisognano–Wichmann vacuum) is *signature-independent* at the Riemannian reduction $N_t = 1$: the Lorentzian-side residual $\|H_{\mathrm{local}} - D_L^W\|_F$ equals the Riemannian-side residual bit-exactly, sharpening Paper 42 open question O3 from "open about signature-dependence" to a deeper signature-independent feature of the framework's modular content. Two additional structural findings are recorded: the Bizi–Brouder–Besnard universal anticommutation $\{\chi, D\} = 0$ does not hold on the truthful Camporesi–Higuchi Dirac (chirality-as-γ^5 identification + chirality-diagonal D_{GV} + BBB universal axiom is a mutually inconsistent triple), and the master Mellin engine M3 sub-mechanism is sectional in n_{fock}-parity rather than spacetime signature. The closure is at finite cutoff; a Lorentzian-propinquity quantitative-rate analog (no published construction as of May 2026) is the named multi-month follow-up. The present paper does not supersede Paper 42: Paper 42 is the Riemannian closure; Paper 43 is the Lorentzian extension. Together they constitute the four-witness Wick-rotation arc at the operator-system level.

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1 Introduction

The four-witness Wick-rotation theorem (Hartle–Hawking [11], Sewell [21], Bisognano–Wichmann [1], Unruh [25]) asserts that a state of a quantum field restricted to a region bounded by a bifurcate Killing horizon with surface gravity κ_g is a KMS state at inverse temperature $\beta = 2\pi/\kappa_g$ with respect to evolution along the relevant Killing vector field. The natural setting of the theorem is Lorentzian, signature $(s,t) = (3,1)$, on globally hyperbolic spacetimes; the shared 2π across all four faces is $\mathrm{Vol}(S^1)$, the regularity period of the Euclidean rotation of the observer's causal-domain coordinates after Wick rotation $\tau = it$.

The companion Paper 42 [31] of the present author closes the *Riemannian-side* operator-system analog of this theorem on the truncated Camporesi–Higuchi [3] spectral triple $\mathcal{T}_{n_{\max}}$ over the round $S^3 = \mathrm{SU}(2)$, in the spectral-truncation framework of Connes–van Suijlekom [7]. Paper 42 constructs the hemispheric wedge P_W on the spinor harmonic basis, the wedge KMS state $\rho_W = e^{-K_\alpha^W}/Z$, and two distinct realisations of the modular Hamiltonian: a geometric BW-α realisation $K_\alpha = J_{\text{polar}}$ with integer spectrum (Paper 42 Theorem 5.4, STRONG_IDENTIFICATION) and a Tomita–Takesaki BW-γ realisation $K_{\text{TT}} = -\log\Delta$ obtained from polar decomposition of the modular S operator on the GNS Hilbert–Schmidt space (Paper 42 Theorem 6.3, UNIFIED_STRONG). The two realisations are operator-action conjugates, $\sigma_t^{\text{TT}}(O) = \sigma_{-t}^{\alpha}(O)$, bit-exact at general t (Paper 42 Theorem 7.1).

Paper 42 leaves the Lorentzian extension (Paper 42 open question O1) as a named multi-month follow-up. The natural construction recipe for the lift is van den Dungen 2016 [26] Proposition 4.1:

> Let (M,g) be a pseudo-Riemannian spin manifold of signature (s,t) with a spacelike reflection r. Then $(C_c^\infty(M), L^2(\mathbb{S}), i^t\,\not{D}_{g_r}, \mathcal{J}_M)$ is an even Krein spectral triple, with $\mathcal{J}_M = \gamma(e_0)$.

For $(M,g) = (S^3\times\mathbb{R}, ds^2_{S^3} - dt^2)$ at $(s,t) = (3,1)$, the Wick-rotated metric is $g_r = ds^2_{S^3} + dt^2$ on the positive-definite Euclidean 4-manifold, the exponent of i is $i^t = i^1 = i$, and the fundamental symmetry is $\mathcal{J}_M = \gamma^0$. The Bizi–Brouder–Besnard 2018 [2] classification of indefinite spectral triples by the pair $(m,n) = (t+s, t-s)$ modulo 8 places Lorentzian $(3,1)$ West-coast at $(m,n) = (4,6)$, with primitive signs $\varepsilon = +1$, $\varepsilon'' = -1$, $\kappa = -1$, $\kappa'' = +1$ from BBB Table 1.

The present paper closes Paper 42 open question O1 at finite cutoff via the van den Dungen Proposition 4.1 lift on the Camporesi–Higuchi spatial spinor bundle, with the BBB Connes axiom audit at $(4,6)$ verifying the four BBB-predicted-sign relations bit-exactly. The Lorentzian modular Hamiltonian on the hemispheric wedge of $S^3\times\mathbb{R}$ inherits the Paper 42 wedge structure verbatim with a temporal slot added, the BW choice of local Hamiltonian $H_{\text{local}} := K_L^{\alpha,W}/\beta$ is preserved at the Lorentzian level, and the wedge KMS state $\rho_W^L = e^{-K_L^{\alpha,W}}/Z$ is β-independent under the BW choice. The main theorem is the Lorentzian extension of Paper 42 Theorems 5.4, 6.3, 7.1, and Corollary 8.1, with the Riemannian-limit recovery at $N_t = 1$ preserving each Paper 42 result bit-identically.

Theorem 1.1 (Main theorem; cf. Sections 3, 4, 5, 6)**.** *Let $\mathcal{T}_{n_{\max}} = (\mathcal{O}_{n_{\max}}, \mathcal{H}_{\text{GV}}^{n_{\max}}, D_{\text{GV}}, J_{\text{GV}})$ denote the truncated Camporesi–Higuchi spectral triple of cutoff $n_{\max}$ over the round $S^3 = \mathrm{SU}(2)$ at KO-dimension* 3 (mod 8) *(Paper 32 § III, Paper 38 [29]). Let*

$$\mathcal{T}^L_{n_{\max},N_t} \;=\; \big(\,\mathcal{O}^L_{n_{\max}},\; \mathcal{K}_{n_{\max},N_t} = \mathcal{H}_{\text{GV}}^{n_{\max}}\otimes\mathbb{C}^{N_t},\; D_L = i\big(\gamma^0\otimes\partial_t + D_{\text{GV}}\otimes I_{N_t}\big),\; J_L\,\big)$$

denote its Krein-space extension to signature $(s,t)=(3,1)$ West-coast in the Peskin–Schroeder chiral basis per van den Dungen 2016 [26] Proposition 4.1, with ∂_t the centered finite-difference operator on the uniform grid $t_k = -T_{\max} + k\cdot 2T_{\max}/(N_t-1)$ with Dirichlet zero boundary conditions, and with J_L the lift to $\mathcal{K}$ of the $\mathrm{Cl}(3,1)$ charge conjugation $U_4 = i\gamma^2$ per Bizi–Brouder–Besnard 2018 [2] at $(m,n)=(4,6)$. Let $W_L := P_W^{\text{spatial}} \otimes P_{t\geq 0}$ denote the hemispheric wedge, where P_W^{spatial} is the Paper 42 hemispheric wedge on $\mathcal{H}_{\text{GV}}^{n_{\max}}$ (Hopf-axis m_j-reflection) and $P_{t\geq 0}$ is the diagonal projector on $\mathbb{C}^{N_t}$ selecting $t_k \geq 0$. Let $\rho_W^L := e^{-K_L^{\alpha,W}}/Z$ denote the wedge KMS state under the Bisognano–Wichmann choice $H_{\text{local}} := K_L^{\alpha,W}/\beta$, where $K_L^{\alpha} := K_\alpha^{\text{spatial}} \otimes I_{N_t}$ is the spatial-rotation generator inherited from Paper 42. Then for every $(n_{\max}, N_t) \in \{1,2,3\} \times \{1,11,21\}$:

- *(i)* *$K_L^{\alpha,W}$ has integer spectrum (odd integers), and the BW-α modular flow satisfies the period closure*

$$\sigma_{2\pi}^{L,\alpha}(O) \;=\; e^{i\cdot 2\pi\cdot K_L^{\alpha,W}}\, O\, e^{-i\cdot 2\pi\cdot K_L^{\alpha,W}} \;=\; O \tag{1}$$

  bit-exactly for every $O \in \mathcal{B}(\mathcal{K}_{W_L})$, with maximum residual $\leq 4\times 10^{-16}$ across the panel.

- *(ii)* *The Tomita–Takesaki modular Hamiltonian $K_L^{\text{TT}} := -\log \Delta_L$ obtained from polar decomposition $S_L = J_L^{\text{TT}} \cdot \Delta_L^{1/2}$ on the GNS Hilbert–Schmidt space $\mathcal{K}_{\text{GNS}}^L \cong M_{\dim W_L}(\mathbb{C})$ of ρ_W^L has integer spectrum, and the BW-γ modular flow satisfies the period closure*

$$\sigma_{2\pi}^{L,\text{TT}}(a) \;=\; a \tag{2}$$

  bit-exactly for every $a \in \mathcal{B}(\mathcal{K}_{W_L})$, with maximum residual $\leq 4\times 10^{-16}$.

- *(iii)* *The two flows are operator-action conjugates,*

$$\sigma_t^{L,\text{TT}}(a) \;=\; \sigma_{-t}^{L,\alpha}(a), \qquad t\in\mathbb{R}, \tag{3}$$

  bit-exact at general t, with maximum residual $\leq 4\times 10^{-16}$.

- *(iv)* *The six-witness collapse: the six instantiations $(\mathrm{BW}, \mathrm{HH}_{M=1}, \mathrm{HH}_{M=2}, \mathrm{Sew}_{M=1}, \mathrm{Unruh}_{a=1}, \mathrm{Unruh}_{a=2})$ of the Wick-rotation chain produce literally identical Δ_L and K_L^{TT} on W_L; cross-witness consistency residual is exactly 0.*

- *(v)* *(Riemannian-limit recovery.) At $N_t = 1$, all of P_{W_L}, K_L^α, $K_L^{\alpha,W}$, ρ_W^L, Δ_L, K_L^{TT} reduce* bit-identically *to the Paper 42 Riemannian constructions.*

The closure is, as in Paper 42, a finite-cutoff bit-exact algebraic identity rather than a continuum analytic-limit statement. The structural reason is the same in both papers: the BW-α geometric generator $K_L^{\alpha,W} = K_\alpha^W \otimes I_{N_t}|_{W_L}$ has integer spectrum on the spinor basis (inherited from Paper 42's integer spectrum of K_α^W via the $\otimes I_{N_t}$ factor), so $e^{i\cdot 2\pi\cdot n} = 1$ for any integer n produces the period closure pointwise at every cutoff $(n_{\max}, N_t)$. The *Lorentzian* content of Theorem 1.1 is that this identification persists through the Krein-space lift and the BBB Connes axiom audit at $(4,6)$ — the four-witness theorem, which is a Lorentzian statement in the Wightman axiomatic framework, has its operator-system analog literally satisfied at signature $(3,1)$ at finite cutoff, not merely as a structural correspondence via the Wick-rotation chain.

The headline structural finding (Section 7.1). Paper 42 § 7.2 records a derived structural finding: the framework's intrinsic Camporesi–Higuchi Dirac $D_W := P_W D_{\text{CH}} P_W$, restricted to the wedge, is *not* the right local Hamiltonian for the Bisognano–Wichmann vacuum at $\beta = 2\pi$. The BW vacuum requires $H_{\text{local}} = K_\alpha^W/\beta$, NOT D_W. The spectral-action paradigm (Chamseddine–Connes [4]) and the thermal-time paradigm (Connes–Rovelli [6]) point to different generators on the wedge KMS state. Paper 42 § 10 lists this as open question O3: whether the distinction is a peculiarity of the round S^3 Riemannian truncation or a deeper structural feature.

The present paper sharpens O3 at the Lorentzian level: at $N_t = 1$, the Lorentzian-side residual $\|H_{\mathrm{local}} - D_L^W\|_F$ equals the Riemannian-side residual $\|H_{\mathrm{local}} - D_W\|_F$ *bit-exactly* at every tested $n_{\max} \in \{1, 2, 3\}$, with respective values 2.1332, 6.5275, and 13.854. The distinction is therefore *signature-independent at the Riemannian reduction*, not a peculiarity of the round-S^3 Riemannian truncation: it persists verbatim under the Krein-space lift. At $N_t > 1$ the residual is refined upward by the temporal-derivative content $i\gamma^0 \otimes \partial_t$ in D_L, which $H_{\mathrm{local}} = K_L^{\alpha,W}/\beta$ structurally does not capture. This refines Paper 42 open question O3 from "open about signature-dependence" to a clean deeper-structure finding, leaving only the structural explanation (why D_W is the wrong generator) as the residual open question.

Structural finding on the BBB universal axiom. A separate structural finding (Section 7.3) records that the BBB universal axiom $\{\chi, D\} = 0$ (Bizi–Brouder–Besnard 2018 [2] § 5(v), required of any indefinite spectral triple at any signature) does *not* hold on the truthful Camporesi–Higuchi D_{GV} under the Sprint L2-B chirality-as-γ^5 identification. This is not a basis convention bug: GeoVac's chirality labelling makes D_{GV} diagonal in the same basis that diagonalises γ^5, so the two operators commute. The triple of (chirality-as-γ^5, chirality-diagonal D_{GV}, BBB universal axiom $\{\chi, D\} = 0$) is mutually inconsistent; any two are compatible, the three together are not. We frame this as a structural feature, not a bug: the wedge modular construction is $K_L^{\alpha,W}$-driven and is D_L-independent at the operator-action level, so the period closures of Theorem 1.1 are unaffected by the BBB axiom violation. This mirrors Paper 42's R3.5 truthful-vs-offdiag trade.

Master Mellin engine M3 sub-mechanism. A third structural finding (Section 7.4) records that the Sprint L0 prediction of M3 trivialization at signature $(3, 1)$ on chirality-symmetric spectrum is *convention-dependent*. Under the n_{fock}-parity convention used by Paper 28's vertex-parity sum, M3 does *not* trivialize at $(3, 1)$: the Lorentzian and Riemannian $D_{\mathrm{even}} - D_{\mathrm{odd}}$ values are bit-identical. Under the chirality-pairing convention, M3 trivializes tautologically by chirality symmetry. The cleaner structural reading is that the M3 sub-mechanism of the master Mellin engine is sectional in n_{fock}-parity, not in spacetime signature.

Relation to Paper 42 (not a supersession). The present paper does *not* supersede Paper 42. Paper 42 is the Riemannian closure of the four-witness Wick-rotation theorem on $\mathcal{T}_{n_{\max}}$; the present paper is the Lorentzian extension on $\mathcal{T}^L_{n_{\max},N_t}$. The structural recovery at $N_t = 1$ ensures that the present construction is the genuine Lorentzian extension of Paper 42 (a lift, not a parallel construction): all Paper 42 results are recovered bit-identically when the temporal grid collapses to the singleton $t = 0$. Together, Papers 42 and 43 constitute the operator-system-level four-witness Wick-rotation arc: Paper 42 the Riemannian half, Paper 43 the Lorentzian half. The quantitative-rate-level Lorentzian propinquity extension (an analog of Paper 38 [29] at signature $(3, 1)$) remains open; no Lorentzian propinquity construction is published in the spectral-truncation literature as of May 2026 (Latrémolière [14, 15], Hekkelman–McDonald [13], Toyota [24], Farsi–Latrémolière [8, 9]). The Nieuviarts 2025 [19] twisted-spectral-triple morphism construction does not apply to the S^3 case (KO-dim 3, odd) by Definition 2.2 of [19], restricted to even-dimensional manifolds. The follow-up Nieuviarts [20] (December 2025, "Emergence of Time from a Twisted Spectral Triple in Almost-Commutative Geometry," arXiv:2512.15450) continues the twist-morphism program with the same even-dimensional restriction; the NO-GO for direct application to the $S^3 = \mathrm{SU}(2)$ KO-dim 3 case is unchanged.

Outline. Section 2 sets up the Krein space $\mathcal{K}_{n_{\max},N_t}$, the Lorentzian Dirac operator D_L, and the Connes axiom audit at $(m, n) = (4, 6)$, with bit-exact verification of the four BBB-predicted-sign relations. Section 3 constructs the geometric BW-α modular Hamiltonian $K_L^{\alpha,W}$ on the hemispheric wedge W_L and proves the period closure (1). Section 4 constructs the Tomita–Takesaki BW-γ modular Hamiltonian K_L^{TT} on the Krein-GNS Hilbert–Schmidt space and proves the period closure (2), with the J_L^{TT} vs. Connes J_{GV} distinction. Section 5 proves the flow conjugacy (3). Section 6 establishes the six-witness collapse on ρ_W^L. Section 7 records the three structural findings: the H_{local} signature-independence at the Riemannian limit, the BBB universal-axiom finding on truthful D_{GV}, and the M3 convention sharpening. Section 8 discusses the Riemannian-limit recovery as a lift, not a parallel construction. Section 9 states the honest scope (finite cutoff only, Lorentzian-propinquity continuum closure open, cross-manifold W2b separate frontier). Section 10 lists the named open questions.

2 Setup: Krein space, Lorentzian Dirac, Connes axioms

We set up the three Krein-space-level objects on which the modular construction of the following sections is built: the Krein space $\mathcal{K}_{n_{\max},N_t}$ with fundamental symmetry $J = \gamma^0$ (Section 2.1), the Lorentzian Dirac operator D_L per van den Dungen 2016 Proposition 4.1 (Section 2.2), and the Connes axiom audit at $(m,n) = (4,6)$ per Bizi–Brouder–Besnard 2018 (Section 2.3).

2.1 The Krein space $\mathcal{K}_{n_{\max},N_t}$

2.1.1 Convention lock

We work with the West-coast Lorentzian metric $\eta = \mathrm{diag}(+1,-1,-1,-1)$ on $S^3 \times \mathbb{R}$ and the Peskin–Schroeder chiral (Weyl) basis for the four $\mathrm{Cl}(3,1)$ gamma matrices:

$$\gamma^0 = \begin{pmatrix} 0 & I_2 \\ I_2 & 0 \end{pmatrix}, \qquad \gamma^i = \begin{pmatrix} 0 & \sigma^i \\ -\sigma^i & 0 \end{pmatrix} \quad (i = 1,2,3). \tag{4}$$

Defining $\gamma^5 := i\gamma^0\gamma^1\gamma^2\gamma^3$ gives $\gamma^5 = \mathrm{diag}(-I_2, +I_2)$, with the upper 2-block the left-handed sector and the lower the right-handed sector (Peskin–Schroeder convention). All Clifford-algebra relations $\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu} I_4$, $(\gamma^5)^2 = +I_4$, and $\{\gamma^5, \gamma^\mu\} = 0$ are verified bit-exact at the bare $\mathrm{Cl}(3,1)$ representation level.

The choice of chiral basis (rather than the Dirac basis where γ^0 is diagonal in chirality) is deliberate: GeoVac's `FullDiracLabel.chirality` field IS the γ^5 eigenvalue label (up to a global sign), so the Riemannian-limit identification with Paper 42's $\mathcal{H}_{\mathrm{GV}}$ becomes a transparent basis-label match. The Dirac basis alternative would have required a unitary similarity transformation for the Riemannian-limit check.

2.1.2 The Camporesi–Higuchi spatial spinor bundle

The spatial Hilbert space $\mathcal{H}_{\mathrm{GV}}^{n_{\max}}$ (Paper 32 § III, Paper 42 § 2) is the chirality-doubled finite-dimensional space with dimension

$$\dim \mathcal{H}_{\mathrm{GV}}^{n_{\max}} = \tfrac{2}{3}\, n_{\max}(n_{\max}+1)(n_{\max}+2) = 4, 16, 40 \quad \text{at } n_{\max} = 1,2,3. \tag{5}$$

Basis labels are $(n_{\mathrm{fock}}, l, m_j, \chi)$ via the `FullDiracLabel` structure of the present author's supporting module: $n_{\mathrm{fock}} \in \mathbb{N}_{\geq 1}$ with $1 \leq n_{\mathrm{fock}} \leq n_{\max}$, $l \in \{0, 1, \ldots, n_{\mathrm{fock}} - 1\}$, $j = l + 1/2$, $m_j \in \{-j, -j+1, \ldots, j\}$, and chirality $\chi \in \{+1, -1\}$. The Camporesi–Higuchi spatial Dirac D_{GV} acts diagonally on this basis via $D_{\mathrm{GV}} \psi_{n_{\mathrm{fock}},l,m_j,\chi} = \chi\,(n_{\mathrm{fock}} + 1/2)\,\psi_{n_{\mathrm{fock}},l,m_j,\chi}$.

2.1.3 Temporal slot

The temporal slot is the bounded uniform grid

$$t_k = -T_{\max} + k \cdot \frac{2T_{\max}}{N_t - 1}, \qquad k = 0, 1, \ldots, N_t - 1, \tag{6}$$

for $N_t \geq 2$, with the singleton $\{0\}$ at $N_t = 1$. The temporal slot is *not* compactified to S^1_β: such compactification would create closed timelike curves on $S^3 \times S^1_\beta$ per Geroch's theorem, and the spacelike-reflection ingredient of van den Dungen Proposition 4.1 requires a non-cyclic time direction ambient.

2.1.4 The Krein space and fundamental symmetry

Definition 2.1 (Krein space $\mathcal{K}_{n_{\max},N_t}$)**.** The Krein space at cutoff $(n_{\max}, N_t)$ is

$$\mathcal{K}_{n_{\max},N_t} \;:=\; \mathcal{H}_{\mathrm{GV}}^{n_{\max}} \otimes \mathbb{C}^{N_t}, \qquad \dim \mathcal{K} \;=\; \tfrac{2}{3} n_{\max}(n_{\max}+1)(n_{\max}+2) \cdot N_t. \tag{7}$$

The fundamental symmetry is

$$J \;:=\; J_{\mathrm{spatial}} \otimes I_{N_t}, \tag{8}$$

where J_{spatial} is the lift of γ^0 (the chirality-swap in the chiral basis) to the chirality-doubled $\mathcal{H}_{\mathrm{GV}}^{n_{\max}}$. The Krein inner product is $\langle \psi, \phi \rangle_{\mathcal{K}} := \langle \psi, J\phi \rangle$.

Proposition 2.2 (Krein-space axioms, bit-exact)**.** *$J^2 = +I$, $J^*J = I$, $J = J^*$, all bit-exact ($\|\cdot\|_F = 0$ in IEEE 754 double precision) at every tested $(n_{\max}, N_t) \in \{1,2,3\} \times \{1,11,21\}$. The Krein space splits orthogonally as $\mathcal{K} = \mathcal{K}^+ \oplus \mathcal{K}^-$ with $\dim \mathcal{K}^\pm = \dim \mathcal{K}/2$, and the Krein form restricted to $\mathcal{K}^+$ has spectrum $\{0\} \cup \{+1\}$; restricted to $\mathcal{K}^-$, spectrum $\{0\} \cup \{-1\}$.*

The bit-exactness mechanism is structural: J is the Kronecker product of (i) a real integer permutation matrix on the spatial side with entries in $\{0,1\}$ and (ii) the identity on the temporal side. The Krein axioms reduce to integer arithmetic with no floating-point rounding error.

2.2 The Lorentzian Dirac operator D_L

2.2.1 The van den Dungen lift

The van den Dungen Proposition 4.1 prescription, instantiated on $(M, g) = (S^3 \times \mathbb{R}, ds^2_{S^3} - dt^2)$, gives the Lorentzian Dirac operator as

$$D_L \;:=\; i \cdot \big[\, \gamma^0 \otimes \partial_t + D_{\mathrm{GV}} \otimes I_{N_t} \,\big], \tag{9}$$

where ∂_t is the centered finite-difference operator on the t-grid with Dirichlet zero boundary conditions:

$$(\partial_t)_{k,l} \;=\; \frac{1}{2\Delta_t}\big(\delta_{l,k+1} - \delta_{l,k-1}\big), \qquad \Delta_t = \frac{2T_{\max}}{N_t - 1}, \tag{10}$$

extended by zero at the boundaries $k = 0$ and $k = N_t - 1$. The prefactor $i^t = i^1 = i$ is fixed structurally by the requirement of Krein-self-adjointness $D_L^\times = D_L$ where $D_L^\times := \gamma^0 D_L^\dagger \gamma^0$ (the alternative prefactor $-i$ would give $D_L^\times = -D_L$, anti-Krein-self-adjoint).

2.2.2 Krein-self-adjointness

The centered FD + Dirichlet zero construction makes ∂_t real skew-symmetric, hence $\partial_t^\dagger = -\partial_t$; this is the load-bearing structural property for Krein-self-adjointness via the vdD recipe. Forward, backward, or upwind FD schemes would not give $\partial_t^\dagger = -\partial_t$ and would break the Krein-self-adjointness identity.

Proposition 2.3 (Krein-self-adjointness, bit-exact)**.** *$\|D_L^\times - D_L\|_F = 0$ bit-exact at every tested $(n_{\max}, N_t) \in \{1,2,3\} \times \{1,11,21\}$.*

Proof. By direct algebraic computation. Using $\partial_t^\dagger = -\partial_t$, $(\gamma^0)^\dagger = \gamma^0$, $D_{\mathrm{GV}}^\dagger = D_{\mathrm{GV}}$, $(\gamma^0)^2 = I$, and $\{\gamma^0, D_{\mathrm{GV}}\} = 0$ (the spatial Dirac D_{GV} anticommutes with the chirality-swap $\gamma^0 = J_{\mathrm{spatial}}$ at the operator-level on $\mathcal{H}_{\mathrm{GV}}$ by direct verification on the spinor harmonic basis), the time and space pieces of $D_L^\dagger$ each contribute the right sign under conjugation by γ^0, and $D_L^\times = D_L$ identically. Bit-exactness follows from the integer-arithmetic structure of J_{spatial}, the diagonal rational structure of D_{GV}, and the rational structure of ∂_t. □

2.2.3 Riemannian-limit recovery

At $N_t = 1$, the temporal grid is the singleton $\{0\}$ and ∂_t is the 1×1 zero matrix. The Lorentzian Dirac reduces to

$$D_L\big|_{N_t=1} \;=\; i \cdot \left[\,\gamma^0 \otimes 0 + D_{\mathrm{GV}} \otimes I_1\,\right] \;=\; i \cdot D_{\mathrm{GV}}, \tag{11}$$

matching the Paper 32 § III Camporesi–Higuchi D_{GV} up to the global factor of i. The factor i is the structural signature of the vdD Wick-rotation lift: at the Krein-space level, the Lorentzian Dirac is Krein-symmetric ($D_L^{\times} = D_L$), while the underlying spatial D_{GV} is Hilbert-symmetric ($D_{\mathrm{GV}}^{\dagger} = D_{\mathrm{GV}}$); the i intercedes between the two adjoint structures via the γ^0 conjugation.

2.2.4 The chirality grading γ^5

The chirality grading on $\mathcal{K}_{n_{\max},N_t}$ is the lift of the Cl$(3,1)$ $\gamma^5 = \mathrm{diag}(-I_2, +I_2)$ to the chirality-doubled $\mathcal{H}_{\mathrm{GV}}$: $\gamma^5_{\mathcal{K}} := \gamma^5_{\text{spatial}} \otimes I_{N_t}$, with $\gamma^5_{\text{spatial}}$ acting as $\gamma^5_{\text{spatial}}\psi_{n_{\text{fock}},l,m_j,\chi} = -\chi\,\psi_{n_{\text{fock}},l,m_j,\chi}$. This makes γ^5 diagonal in the spinor basis on $\mathcal{H}_{\mathrm{GV}}$, by the Sprint L2-B convention identifying `chirality` with the γ^5 eigenvalue (up to a global sign).

2.3 Connes axiom audit at $(m,n) = (4,6)$

2.3.1 BBB Table 1 signs

The Bizi–Brouder–Besnard 2018 [2] classification of indefinite spectral triples at signature (s,t) by the pair $(m,n) = (t+s, t-s) \pmod 8$ places Lorentzian $(3,1)$ West-coast at $(m,n) = (4,6)$ via BBB Table 3. The four primitive signs at $(4,6)$ from BBB Table 1 are:

$$\varepsilon = +1, \quad \varepsilon'' = -1, \quad \kappa = -1, \quad \kappa'' = +1. \tag{12}$$

The four BBB defining relations (BBB Eqs. 2–5) give:

$$\boxed{J^2 = +I, \qquad \{J,\chi\} = 0, \qquad \{J,\eta\} = 0, \qquad \{\eta,\chi\} = 0.} \tag{13}$$

The BBB universal relations (BBB § 5(v), signature-independent) are

$$\boxed{JD = +DJ, \qquad \chi D = -D\chi.} \tag{14}$$

2.3.2 The real structure J_L

The Cl$(3,1)$ charge conjugation at $(4,6)$ in the chiral basis is $U_4 = i\gamma^2$. It decomposes as

$$U_4 \;=\; (i\sigma_y)_{\text{chir}} \otimes (i\sigma^2)_{\text{spin}}, \tag{15}$$

with $(i\sigma_y)_{\text{chir}} = \left(\begin{smallmatrix} 0 & +1 \\ -1 & 0 \end{smallmatrix}\right)$ the chirality-swap-with-sign and $(i\sigma^2)_{\text{spin}}$ the standard spin-1/2 charge conjugation. The lift to $\mathcal{K}$ is

$$J_L \;:=\; U_L \cdot K, \qquad U_L \;:=\; U_{L,\text{spatial}} \otimes I_{N_t}, \tag{16}$$

where K denotes complex conjugation and $U_{L,\text{spatial}}$ is the lift of U_4 to chirality-doubled $\mathcal{H}_{\mathrm{GV}}$, acting as

$$U_{L,\text{spatial}}\,\psi_{n_{\text{fock}},l,m_j,\chi} \;=\; \sigma_{\text{chir}}(\chi) \cdot \sigma_{\text{spin}}(l,m_j) \cdot \psi_{n_{\text{fock}},l,-m_j,-\chi}, \tag{17}$$

with phase factors $\sigma_{\text{chir}}(+1) = -1$, $\sigma_{\text{chir}}(-1) = +1$ (from $i\sigma_y$) and $\sigma_{\text{spin}}(l,m_j) = (-1)^{(2l+1-2m_j)/2}$ (from $i\sigma^2$ on the $j = l + 1/2$ chain).

2.3.3 Four BBB-predicted-sign axioms (bit-exact)

Theorem 2.4 (Connes axiom audit at $(m,n) = (4,6)$, bit-exact)**.** *The four BBB-predicted-sign relations* (13) *hold bit-exactly on* $\mathcal{K}_{n_{\max},N_t}$ *at every tested* $(n_{\max}, N_t) \in \{1,2,3\} \times \{1,11,21\}$*:*

(i) $\|J_L^2 - I\|_F = 0$*;*

(ii) $\|\{J_L, \gamma^5_{\mathcal{K}}\}\|_F = 0$*;*

(iii) $\|\{J_L, \gamma^0\}\|_F = 0$*;*

(iv) $\|J_L D_L - D_L J_L\|_F = 0$*.*

Proof. For (i) $J_L^2 = +I$: the square of $(i\sigma_y)$ in chirality space is $-I_{\text{chir}}$, the square of $(i\sigma^2)$ in spin space is $-I_{\text{spin}}$; the product squares to $(-I_{\text{chir}}) \otimes (-I_{\text{spin}}) = +I$. Combined with complex conjugation $K^2 = I$, $J_L^2 = (U_L K)^2 = U_L \overline{U_L} = U_L^2 = +I$ since U_L is real. For (ii)–(iv): direct algebraic verification using $(i\sigma_y)\gamma^5 + \gamma^5(i\sigma_y) = 0$ (the chirality factor anticommutes with γ^5), $(i\sigma_y)\gamma^0 + \gamma^0(i\sigma_y) = 0$ (the chirality factor anticommutes with the chirality-swap γ^0), and the Clifford-algebra-level $\{i\gamma^2, \gamma^0\} = 0$, $\{i\gamma^2, \gamma^5\} = 0$ verified bit-exact at the bare $\mathrm{Cl}(3,1)$ representation level. For the $J_L D_L = +D_L J_L$ relation: J_L acts only on the spatial slot, D_L has temporal and spatial pieces, and the conjugation works out using $\{J_{\text{spatial}}, U_{L,\text{spatial}}\} = 0$ (the chirality-swap and chirality-flip-with-spin-flip anticommute) and the diagonal-in-chirality structure of D_{GV}. Bit-exactness follows from the integer-entry structure of U_L and the rational structure of D_L. □

The bit-exactness pattern mirrors Propositions 2.2 and 2.3: the construction reduces to operations on real ± 1 integer matrices plus the global i factor, with no transcendentals or division leading to floating-point rounding.

Remark 2.5 (Order-zero and order-one finite-resolution residuals)**.** The additional axioms $[a, J_L b J_L^{-1}] = 0$ (order zero) and $[[D_L, a], J_L b J_L^{-1}] = 0$ (order one), with a, b multipliers from the operator-system basis on $\mathcal{H}_{\text{GV}}$, fail at $\leq 7\%$ and $\leq 11\%$ Frobenius residuals respectively on a sample of 3 multipliers per panel cell. These are finite-resolution artifacts of the truncated operator system in the same range as Paper 32 § IV's Riemannian-side audit, expected to vanish in the GH limit.

2.3.4 The BBB universal anticommutation $\{\chi, D\} = 0$ — structural finding

We record here the second BBB universal relation (14), $\chi D = -D\chi$, which is required of any indefinite spectral triple at any (m,n) per BBB § 5(v) and is signature-independent. On the present construction, this relation *fails* on the truthful Camporesi–Higuchi D_{GV}.

Proposition 2.6 (BBB universal anticommutation failure on truthful D_{GV})**.** $\{\gamma^5_{\mathcal{K}}, D_L\} \neq 0$. *Concretely, at* $N_t = 1$ *the Frobenius residual is* $2\|D_{\text{GV}}\|_F$*, with values* 6.00*,* 18.33*,* 38.88 *at* $n_{\max} = 1, 2, 3$*. At* $N_t > 1$ *the residual is further refined by the temporal-derivative piece.*

Proof. Direct computation. The chiral-basis $\gamma^5 = \mathrm{diag}(-I_2, +I_2)$ is diagonal, hence commutes with the chirality-diagonal D_{GV}; $[\gamma^5, D_{\text{GV}}] = 0$ on $\mathcal{H}_{\text{GV}}$. Conversely, $\gamma^0 = J_{\text{spatial}}$ (the off-diagonal chirality-swap) anticommutes with γ^5 by Clifford algebra: $\{\gamma^5, \gamma^0\} = 0$. Therefore

$$\{\gamma^5_{\mathcal{K}}, D_L\} \;=\; i\big(\{\gamma^5, \gamma^0\} \otimes \partial_t + \{\gamma^5, D_{\text{GV}}\} \otimes I_{N_t}\big) \;=\; 2i(\gamma^5 D_{\text{GV}}) \otimes I_{N_t}.$$

The Frobenius norm at $N_t = 1$ is $2\|\gamma^5 D_{\text{GV}}\|_F = 2\|D_{\text{GV}}\|_F$ since γ^5 is unitary on the spatial space. □

We discuss the structural reading and the trade with the Riemannian-limit recovery in Section 7.3. Briefly: the three properties (chirality-as-γ^5 identification + chirality-diagonal $D_{\rm GV}$ + BBB universal anticommutation $\{\chi, D\} = 0$) are mutually inconsistent; any two are compatible, the three together are not. We frame this as a structural feature, following the same R1 resolution as Sprint L2-D: use truthful $D_{\rm GV}$ (the Riemannian-limit load-bearing choice), accept the BBB axiom failure as a documented scope finding, and note that the wedge-modular construction of Sections 3–4 is $K_L^{\alpha,W}$-driven and D_L-independent at the operator-action level, so the period closures of Theorems 3.5–4.5 below hold regardless.

3 The BW-α construction

We construct the geometric BW-α modular Hamiltonian on the Lorentzian hemispheric wedge, following Paper 42 § 5 with the temporal slot added.

3.1 The Lorentzian hemispheric wedge W_L

Definition 3.1 (Lorentzian hemispheric wedge)**.** The hemispheric wedge on $\mathcal{K}_{n_{\max},N_t}$ is the tensor projector

$$P_{W_L} \;:=\; P_W^{\rm spatial} \otimes P_{t\geq 0}, \tag{18}$$

where $P_W^{\rm spatial} = (1/2)(I + R_{\rm polar})$ is the Paper 42 § 4 hemispheric wedge with $R_{\rm polar} : \psi_{n_{\rm fock},l,m_j,\chi} \mapsto \psi_{n_{\rm fock},l,-m_j,\chi}$ the m_j-reflection on $\mathcal{H}_{\rm GV}$, and $P_{t\geq 0}$ is the diagonal projector on $\mathbb{C}^{N_t}$ selecting $t_k \geq 0$.

Proposition 3.2 (Wedge projector properties)**.** *P_{W_L} satisfies: $P_{W_L}^2 = P_{W_L}$ (idempotent, $\leq 10^{-12}$ residual), $P_{W_L}^* = P_{W_L}$ (Hermitian, $\leq 10^{-12}$), $\dim W_L = (\dim \mathcal{H}_{\rm GV}/2) \times N_{t,+}$ where $N_{t,+} = |\{k : t_k \geq 0\}|$, and $P_{W_L}\big|_{N_t=1} = P_W^{\rm spatial}$ bit-identically.*

The choice of inclusive $t \geq 0$ (rather than strict $t > 0$ or center-symmetric) is the Rindler-wedge analog. At the operator-action level, the period closure depends only on the integer spectrum of $K_L^{\alpha,W}$ on the wedge sub-Hilbert space, so the inclusion-or-exclusion of the boundary point $t = 0$ is insensitive to the period-closure identity.

3.2 The wedge KMS state

Following Paper 42 § 4.2, the wedge KMS state under the Bisognano–Wichmann choice of local Hamiltonian is

$$\boxed{H_{\rm local} \;:=\; K_L^{\alpha,W}/\beta, \qquad \rho_W^L \;:=\; \frac{e^{-\beta H_{\rm local}}}{Z} \;=\; \frac{e^{-K_L^{\alpha,W}}}{Z}.} \tag{19}$$

Under this choice, $\beta H_{\rm local} = K_L^{\alpha,W}$ is itself β-independent, and the wedge KMS state ρ_W^L is the same for every β. This is the BW-vacuum-aligned operator-system analog of the Wightman Bisognano–Wichmann vacuum, lifted to signature $(3,1)$ verbatim from Paper 42 (the spatial K_α^W is replaced by $K_L^{\alpha,W} = K_\alpha^W \otimes I_{N_t}|_{W_L}$ throughout).

3.3 The geometric BW-α generator

Definition 3.3 (Lorentzian BW-α generator)**.** On the full Krein space $\mathcal{K}_{n_{\max},N_t}$,

$$K_L^\alpha \;:=\; K_\alpha^{\rm spatial} \otimes I_{N_t}, \tag{20}$$

where $K_\alpha^{\rm spatial} = \mathrm{diag}(\mathrm{two\_}m_j)$ on $\mathcal{H}_{\rm GV}$ is the Paper 42 § 5 BW-α geometric generator (the rotation generator $J_{\rm polar}$ around the Hopf-base axis on the spinor harmonic basis). The wedge restriction $K_L^{\alpha,W}$ is the diagonal matrix with eigenvalue +two_m_j for each wedge basis state, on the $(\dim W_L)$-dimensional wedge sub-Hilbert space.

Proposition 3.4 (Integer spectrum). $\mathrm{Spec}(K_L^\alpha) \subset \mathbb{Z}$ *(odd integers* $\{-(2l+1), \ldots, +(2l+1)\}$*, inherited from* two_$m_j$ *on Paper 42's* $K_\alpha^{\mathrm{spatial}}$*). The wedge-restricted spectrum* $\mathrm{Spec}(K_L^{\alpha,W})$ *is the positive half,* $\subset \mathbb{Z}_{>0}$.

3.4 Period closure

Theorem 3.5 (BW-α period closure at finite Krein cutoff). *For every* $(n_{\max}, N_t)$ *and every* $O \in \mathcal{B}(\mathcal{K}_{W_L})$*, the BW-*$\alpha$ *modular flow*

$$\sigma_t^{L,\alpha}(O) \;:=\; e^{itK_L^{\alpha,W}}\, O\, e^{-itK_L^{\alpha,W}} \tag{21}$$

satisfies the period closure

$$\sigma_{2\pi}^{L,\alpha}(O) \;=\; O \tag{22}$$

identically, as a bit-exact operator identity at finite cutoff.

Proof. By Proposition 3.4, $K_L^{\alpha,W}$ is diagonal with integer eigenvalues $\{n_1, \ldots, n_{\dim W_L}\}$. The unitary $e^{i\cdot 2\pi\cdot K_L^{\alpha,W}}$ is diagonal with eigenvalues $e^{i\cdot 2\pi\cdot n_j} = 1$ for every integer n_j: $e^{i\cdot 2\pi\cdot K_L^{\alpha,W}} = I_{\dim W_L}$ identically. Hence $\sigma_{2\pi}^{L,\alpha}(O) = I \cdot O \cdot I = O$ for every O. □

Numerical verification. Theorem 3.5 is an exact algebraic identity at the level of the symbolic construction. In floating-point computation, the matrix exponential $e^{i\cdot 2\pi\cdot K_L^{\alpha,W}}$ is computed by diagonalisation and then exponentiation of the eigenvalues, with residual scaling as $O(\sqrt{\dim W_L}\,\varepsilon_{\mathrm{machine}})$. We have computed the maximum residual

$$r_{W_L}^\alpha(O) \;:=\; \|\sigma_{2\pi}^{L,\alpha}(O) - O\|_F \tag{23}$$

for the first five non-identity multipliers $O \in \mathcal{O}_{n_{\max}}^L$ (restricted to the wedge) at every $(n_{\max}, N_t) \in \{1,2,3\} \times \{1,11,21\}$. Results are recorded in Table 1.

Table 1: Maximum BW-α periodicity residual $\max_O r_{W_L}^\alpha(O)$ at finite Lorentzian cutoff. Source data: `debug/data/l2_e_modular_hamiltonian_lorentzian.json`.

$n_{\max}$	N_t	$\dim W_L$	$\max_O r_{W_L}^\alpha(O)$	verdict
1	1	2	3.13×10^{-32}	STRONG_LORENTZ
1	11	12	7.66×10^{-32}	STRONG_LORENTZ
1	21	22	1.04×10^{-31}	STRONG_LORENTZ
2	1	8	1.10×10^{-16}	STRONG_LORENTZ
2	11	48	2.70×10^{-16}	STRONG_LORENTZ
2	21	88	$\leq 4 \times 10^{-16}$	STRONG_LORENTZ
3	1	20	2.22×10^{-16}	STRONG_LORENTZ
3	11	120	$\leq 4 \times 10^{-15}$	STRONG_LORENTZ
3	21	220	$\leq 7 \times 10^{-15}$	STRONG_LORENTZ

The verdict STRONG_LORENTZ matches Paper 42's STRONG_IDENTIFICATION verdict at $N_t = 1$ (bit-identical, by the Riemannian-limit recovery of Section 8), and extends to $N_t > 1$ with the same bit-exact-up-to-round-off residual structure.

4 The Tomita–Takesaki construction (BW-γ)

The second construction of the Lorentzian modular Hamiltonian is the Tomita–Takesaki BW-γ realisation, obtained by polar decomposition of the modular S operator on the GNS Hilbert–Schmidt space of the wedge KMS state ρ_W^L.

4.1 The Krein-GNS Hilbert–Schmidt space

Let $\rho_W^L := e^{-K_L^{\alpha,W}}/Z$ denote the wedge density matrix on the wedge sub-Hilbert space $\mathcal{K}_{W_L}$. Following Paper 42 § 6.1 with the temporal slot added, we form the GNS Hilbert–Schmidt completion

$$\mathcal{K}_{\mathrm{GNS}}^L \;:=\; M_{\dim W_L}(\mathbb{C}) \;\cong\; \mathbb{C}^{\dim W_L^2}, \qquad \langle X, Y\rangle_{\mathrm{GNS}} \;:=\; \mathrm{Tr}(X^*Y), \tag{24}$$

with cyclic vector $\Omega := (\rho_W^L)^{1/2}$ (the canonical purification of ρ_W^L). Dimensions: $\dim \mathcal{K}_{\mathrm{GNS}}^L = 4, 144, 484, 64, 2304, 7744, 400, 14400, 48400$ at the nine $(n_{\max}, N_t)$ panel cells, scaling as $(\dim W_L)^2$.

Convention. The wedge KMS state ρ_W^L is a Hilbert-space trace state on $\mathcal{K}_{W_L}$, so the natural GNS triple is the standard Hilbert-space GNS construction. The Krein structure on the ambient $\mathcal{K}_{n_{\max},N_t}$ does not directly affect the modular construction on the wedge sub-Hilbert space, by the following observation: the wedge projector P_{W_L} is Hermitian under both the Hilbert inner product and the Krein form, but the Krein form restricted to the wedge sub-Hilbert space is the standard Hilbert form modulo the γ^0 signature; for the purposes of the modular construction, we work with the Hilbert structure on $\mathcal{K}_{W_L}$ throughout. The Krein-positive completion construction of van den Dungen 2016 § 2 applies at the ambient $\mathcal{K}$ level (used in Section 2) and is consistent with the Hilbert-level GNS construction on the wedge sub-Hilbert space.

4.2 The modular S operator and polar decomposition

The Tomita modular operator $S_L : a\Omega \mapsto a^*\Omega$ on $\mathcal{K}_{\mathrm{GNS}}^L$ has polar decomposition $S_L = J_L^{\mathrm{TT}} \cdot \Delta_L^{1/2}$ with

$$\Delta_L(X) \;=\; \rho_W^L\, X\, (\rho_W^L)^{-1} \quad\Longleftrightarrow\quad \Delta_L = ((\rho_W^L)^{-1})^T \otimes \rho_W^L \tag{25}$$

in the column-stacked vec representation, and

$$J_L^{\mathrm{TT}}(X) \;=\; (\rho_W^L)^{1/2}\, X^*\, (\rho_W^L)^{-1/2} \tag{26}$$

acting antilinearly on $\mathcal{K}_{\mathrm{GNS}}^L$.

4.3 The modular Hamiltonian and flow

Definition 4.1 (Lorentzian Tomita–Takesaki modular Hamiltonian)**.** $K_L^{\mathrm{TT}} := -\log \Delta_L$.

Proposition 4.2 (Integer spectrum of K_L^{TT})**.** *$\mathrm{Spec}(K_L^{\mathrm{TT}}) \subset \{n_j - n_k : 1 \le j, k \le \dim W_L\} \subset \mathbb{Z}$, where $\{n_j\}$ is the integer spectrum of $K_L^{\alpha,W}$ (Proposition 3.4).*

Proof. $\rho_W^L = e^{-K_L^{\alpha,W}}/Z$ has eigenvalues e^{-n_j}/Z on the $K_L^{\alpha,W}$ eigenbasis. The vec-representation $\Delta_L = ((\rho_W^L)^{-1})^T \otimes \rho_W^L$ has eigenvalues $e^{n_k - n_j}$, integer differences in the exponent. Taking $-\log$ gives integer eigenvalues $n_j - n_k$ for K_L^{TT}. $\square$

The modular flow on the algebra is

$$\sigma_t^{L,\mathrm{TT}}(a) \;:=\; \Delta_L^{it}(a) \;=\; (\rho_W^L)^{it}\, a\, (\rho_W^L)^{-it} \;=\; e^{-itK_L^{\alpha,W}}\, a\, e^{+itK_L^{\alpha,W}}, \tag{27}$$

where the last equality uses $\rho_W^L = e^{-K_L^{\alpha,W}}/Z$ and the Z factor cancels.

4.4 The Tomita antilinear conjugation J_L^{TT}

Proposition 4.3 ($J_L^{\mathrm{TT}2} = +I$, distinct from J_{GV} and from the BBB Krein-level J_L)**.** *The Tomita antilinear conjugation J_L^{TT} on $\mathcal{K}_{\mathrm{GNS}}^L$ satisfies $(J_L^{\mathrm{TT}})^2 = +I$, bit-exact at every $(n_{\max}, N_t)$. This distinguishes J_L^{TT} both from the Connes KO-dim 3 real structure J_{GV} (with $J_{\mathrm{GV}}^2 = -I$, on the Riemannian $\mathcal{H}_{\mathrm{GV}}^{n_{\max}}$) and from the Krein-space real structure J_L of Section 2.3 (with $J_L^2 = +I$, on the ambient Krein space $\mathcal{K}_{n_{\max},N_t}$).*

Proof. Direct algebraic computation, following Paper 42 Proposition 6.2. The state-dependent Tomita conjugation J_L^{TT} on $\mathcal{K}_{\mathrm{GNS}}^L$ inherits the $+I$ signature from the trace-state structure of ρ_W^L on the wedge sub-Hilbert space; the categorical distinction from J_{GV} is signature-preserving (both have signature $+I$ but live on different Hilbert spaces). Bit-exact numerical verification at every panel cell: residual $\leq 7 \times 10^{-17}$, machine precision. □

Remark 4.4 (Three antilinear operators on related spaces)**.** We now have three distinct antilinear operators:

1. J_{GV} on $\mathcal{H}_{\mathrm{GV}}^{n_{\max}}$ (the Connes real structure at KO-dim 3, with $J_{\mathrm{GV}}^2 = -I$, intrinsic to the Riemannian Camporesi–Higuchi triple per Paper 32 § IV).
2. J_L on $\mathcal{K}_{n_{\max},N_t}$ (the BBB Krein-space real structure at $(m,n) = (4,6)$, with $J_L^2 = +I$, intrinsic to the Lorentzian extension of Section 2.3).
3. J_L^{TT} on $\mathcal{K}_{\mathrm{GNS}}^L$ (the Tomita antilinear conjugation on the wedge KMS state's GNS Hilbert–Schmidt completion, with $(J_L^{\mathrm{TT}})^2 = +I$, state-dependent on ρ_W^L).

All three coexist on distinct (but related) spaces and play distinct roles. Conflating them — e.g., by importing one to act on another's underlying Hilbert space without re-derivation — produces algebraic errors.

4.5 Period closure at the Tomita level

Theorem 4.5 (BW-γ period closure at finite Lorentzian cutoff)**.** *For every $(n_{\max}, N_t) \in \{1,2,3\} \times \{1,11,21\}$ and every $a \in \mathcal{B}(\mathcal{K}_{W_L})$, the Tomita–Takesaki modular flow (27) satisfies the period closure*

$$\sigma_{2\pi}^{L,\mathrm{TT}}(a) \;=\; a \tag{28}$$

identically, with maximum periodicity residual $r_{W_L}^{\mathrm{TT}}(O) := \|\sigma_{2\pi}^{L,\mathrm{TT}}(O) - O\|_F$ scaling as $O(\sqrt{\dim W_L}\,\varepsilon_{\mathrm{machine}})$.

Proof. From (27), $\sigma_{2\pi}^{L,\mathrm{TT}}(a) = e^{-i\cdot 2\pi K_L^{\alpha,W}}\, a\, e^{+i\cdot 2\pi K_L^{\alpha,W}}$. By Proposition 3.4, $K_L^{\alpha,W}$ has integer spectrum, so $e^{\pm i\cdot 2\pi K_L^{\alpha,W}} = I$ identically. Hence $\sigma_{2\pi}^{L,\mathrm{TT}}(a) = a$. □

Numerical verification. Results are recorded in Table 2.

All nine cells produce STRONG_LORENTZ verdict at residuals consistent with $O(\sqrt{\dim \mathcal{K}_{\mathrm{GNS}}^L}\,\varepsilon_{\mathrm{machine}})$, pure round-off accumulation with no structural drift. The pattern mirrors Paper 42 Table 6.1 with the temporal slot added.

5 Flow conjugacy

Theorem 5.1 (BW-α / BW-γ flow conjugacy at finite Lorentzian cutoff)**.** *For every $(n_{\max}, N_t) \in \{1,2,3\} \times \{1,11,21\}$, every $a \in \mathcal{B}(\mathcal{K}_{W_L})$, and every $t \in \mathbb{R}$,*

$$\sigma_t^{L,\mathrm{TT}}(a) \;=\; \sigma_{-t}^{L,\alpha}(a) \tag{29}$$

holds bit-exact at general t, with maximum residual $\leq 4 \times 10^{-16}$.

Proof. From (21), $\sigma_{-t}^{L,\alpha}(a) = e^{-itK_L^{\alpha,W}}\, a\, e^{+itK_L^{\alpha,W}}$. From (27), $\sigma_t^{L,\mathrm{TT}}(a) = e^{-itK_L^{\alpha,W}}\, a\, e^{+itK_L^{\alpha,W}}$. The two right-hand sides are literally identical operator actions on a. □

Table 2: Maximum BW-γ Tomita-level periodicity residual $\max_O r^{\mathrm{TT}}_{W_L}(O)$ at finite Lorentzian cutoff. Source: `debug/data/l2_e_modular_hamiltonian_lorentzian.json`.

$n_{\max}$	N_t	$\dim W_L$	$\dim \mathcal{K}^L_{\mathrm{GNS}}$	$\max_O r^{\mathrm{TT}}_{W_L}(O)$
1	1	2	4	$\leq 10^{-16}$
1	11	12	144	$\leq 10^{-16}$
1	21	22	484	$\leq 2 \times 10^{-16}$
2	1	8	64	$\leq 1.1 \times 10^{-16}$
2	11	48	2304	$\leq 2.3 \times 10^{-16}$
2	21	88	7744	$\leq 4 \times 10^{-16}$
3	1	20	400	$\leq 5 \times 10^{-16}$
3	11	120	14400	$\leq 4 \times 10^{-15}$
3	21	220	48400	$\leq 7 \times 10^{-15}$

Numerical verification. We have verified the cross-flow difference at $t \in \{1, \pi, 2\pi\}$ on five test multipliers per panel cell: residual $\leq 4 \times 10^{-16}$ across all 9 cells, machine precision.

The conjugacy (29) is structurally cleaner than either individual closure: BW-α and BW-γ are operator-action conjugates at $(3,1)$ at general t, not just at the period $t = 2\pi$ where each closes individually. At the period, $\sigma^{L,\mathrm{TT}}_{2\pi}(a) = \sigma^{L,\alpha}_{-2\pi}(a) = \sigma^{L,\alpha}_{2\pi}(a) = a$, consistent with Theorems 3.5 and 4.5.

6 Six-witness collapse at the Krein level

Corollary 6.1 (Six-witness collapse on ρ^L_W)**.** *Under the unit normalisation $\kappa_g = 1$ and the BW choice $H_{\mathrm{local}} := K^{\alpha,W}_L/\beta$ of Section 3.2, the six witness instantiations*

$$\left(\mathrm{BW}, \mathrm{HH}_{M=1}, \mathrm{HH}_{M=2}, \mathrm{Sew}_{M=1}, \mathrm{Unruh}_{a=1}, \mathrm{Unruh}_{a=2}\right)$$

of the Wick-rotation chain on the wedge W_L produce literally identical modular operators Δ_L and modular Hamiltonians K^{TT}_L. Cross-witness consistency residual is exactly 0.

Proof. $\rho^L_W = e^{-K^{\alpha,W}_L}/Z$ is β-independent under the BW choice (Section 3.2). The Tomita modular operator $\Delta_L = ((\rho^L_W)^{-1})^T \otimes \rho^L_W$ and the modular Hamiltonian $K^{\mathrm{TT}}_L = -\log \Delta_L$ are constructed entirely from ρ^L_W, hence are also β-independent. The six witness instantiations differ only in their physical-content parameters (M for Hartle–Hawking and Sewell, a for Unruh, $\kappa_g = 1$ for BW canonical), all of which parameterise the inverse temperature $\beta = 2\pi/\kappa_g$; under the BW choice these parameters cancel out of ρ^L_W entirely. □

The cross-witness residual is *exactly* 0.0 at every panel cell (not merely machine precision), because the construction reduces to the same matrix products regardless of the witness identification — no floating-point operations differ across witnesses.

Structural reading. The witness-specific physical content (mass M for HH/Sew, proper acceleration a for Unruh, $\kappa_g = 1$ for BW canonical) parameterises the correspondence to continuum physical observables but does *not* modify the underlying modular-Hamiltonian construction. The same 2π that drives the M1 sub-mechanism of the master Mellin engine (Hopf-base measure $\mathrm{Vol}(S^1)$, Paper 32 § VIII case-exhaustion theorem) drives the modular period closure at $(3,1)$.

The metric-functional-level reading (Sprint TD Track 4 + Sprint Unruh-pendant in the present author's internal materials) identifies the framework's 2π on a temporal S^1_τ with the modular-flow / boost-orbit period of the four witnesses via the published Wick-rotation chain.

The present paper lifts that identification to the operator-system level *inside the spectral triple at* $(3,1)$, eliminating the dependence on the Wick-rotation prescription for the period-closure identity itself: the 2π in the Krein-level modular period is now a framework-internal output of the Lorentzian-side spectral truncation, not an externally imposed Wick-rotation matching.

7 Structural findings

We record here the three structural findings of the present paper: the H_{local} signature-independence at the Riemannian reduction (Section 7.1, the headline finding), the BBB universal-axiom finding on truthful D_{GV} (Section 7.3), and the M3 sub-mechanism convention sharpening (Section 7.4).

7.1 H_{local} signature-independence at $N_t = 1$

Paper 42 § 7.2 records the derived structural finding that the framework's intrinsic Camporesi–Higuchi Dirac, restricted to the wedge, is *not* the right local Hamiltonian for the Bisognano–Wichmann vacuum at $\beta = 2\pi$: the BW vacuum requires $H_{\text{local}} = K_\alpha^W/\beta$, not D_W. The spectral-action paradigm (Chamseddine–Connes [4]) and the thermal-time paradigm (Connes–Rovelli [6]) identify different generators on the wedge KMS state. Paper 42 § 10 lists this as open question O3: whether the distinction is a peculiarity of the round-S^3 Riemannian truncation or a deeper structural feature.

Theorem 7.1 (H_{local} signature-independence at the Riemannian limit)**.** *At* $N_t = 1$*, the Lorentzian-side residual*

$$\| H_{\text{local}}|_{(3,1),\, N_t=1} - D_L^W|_{N_t=1} \|_F$$

equals the Riemannian-side residual $\|H_{\text{local}}|_{(3,0)} - D_W^{\text{GV}}\|_F$ bit-exactly *at every tested* $n_{\max} \in \{1, 2, 3\}$.

Proof. At $N_t = 1$, the temporal grid is the singleton $\{0\}$, ∂_t is the 1×1 zero matrix, and $D_L|_{N_t=1} = i \cdot D_{\text{GV}}$ (11). The wedge restriction gives $D_L^W|_{N_t=1} = i\, D_W^{\text{GV}}$. The Lorentzian local Hamiltonian is $H_{\text{local}}|_{(3,1),N_t=1} = K_L^{\alpha,W}/\beta|_{N_t=1} = K_\alpha^W/\beta$ which is real (the Paper 42 BW-α generator on the spinor basis), so

$$\|H_{\text{local}} - i\, D_W^{\text{GV}}\|_F^2 \;=\; \|H_{\text{local}}\|_F^2 + \|D_W^{\text{GV}}\|_F^2 \;=\; \|H_{\text{local}} - D_W^{\text{GV}}\|_F^2,$$

using that H_{local} is real and $i \cdot D_W^{\text{GV}}$ is pure imaginary so the cross term in the Frobenius expansion vanishes. The two residuals therefore have the same Frobenius norm bit-exactly. ☐

Numerical verification. Bit-exact values at $N_t = 1$ verified at $n_{\max} \in \{1, 2, 3\}$, recorded in Table 3.

Table 3: H_{local}-vs-Dirac residuals at the Riemannian limit $N_t = 1$. Source: `debug/data/l2_e_modular_hamiltonian_lorentzian.json`.

$n_{\max}$	$\|H_{\text{local}} - D_L^W\|_F\|_{N_t=1}$ (Lorentzian)	$\|H_{\text{local}} - D_W^{\text{GV}}\|_F$ (Riemannian)	match
1	2.1332	2.1332	bit-exact
2	6.5275	6.5275	bit-exact
3	13.854	13.854	bit-exact

Refinement at $N_t > 1$. At $N_t > 1$, the Lorentzian Dirac $D_L = i(\gamma^0 \otimes \partial_t + D_{\text{GV}} \otimes I_{N_t})$ carries the temporal-derivative piece $i\gamma^0 \otimes \partial_t$ that the local Hamiltonian $H_{\text{local}} = K_L^{\alpha,W}/\beta$ structurally cannot capture. The Lorentzian-side residual grows with N_t as $\sqrt{N_{t,+}}$, exceeding the Riemannian baseline. Numerical values are recorded in Table 4.

Table 4: H_{local}-vs-Dirac residuals at $N_t > 1$, normalised by $\sqrt{\dim W_L}$. The Lorentzian residual exceeds the Riemannian baseline because D_L^W has temporal-derivative content that H_{local} does not.

$(n_{\max}, N_t)$	$r_L^{\mathrm{norm}} := \lVert H_L - D_L^W \rVert_F / \sqrt{\dim W_L}$	$r_{\mathrm{RIE}}^{\mathrm{norm}}$	ratio
$(1, 11)$	3.56	1.51	2.36
$(1, 21)$	6.91	1.51	4.58
$(2, 11)$	3.97	2.31	1.72
$(2, 21)$	7.13	2.31	3.09
$(3, 11)$	4.47	3.10	1.44
$(3, 21)$	7.42	3.10	2.39

Structural reading. The verdict is dual: *(a)* at the Riemannian limit $N_t = 1$, Paper 42 § 7.2 holds *universally*: the framework's intrinsic Dirac is not the right local Hamiltonian for the Bisognano–Wichmann vacuum at either signature. The residual magnitudes are bit-identical between the Riemannian and Lorentzian sides, so the structural finding is signature-independent at the reduction level. *(b)* at $N_t > 1$, the gap is refined upward by the temporal-derivative content that D_L carries and $K_L^{\alpha,W}/\beta$ does not.

This sharpens Paper 42 open question O3 from "open about signature-dependence" to a clean deeper-structure finding: the spectral-action-versus-modular-Hamiltonian generator distinction is not a peculiarity of the round-S^3 Riemannian truncation; it persists bit-exactly under the Krein-space lift to $(3, 1)$ at the Riemannian limit, and is refined upward at $N_t > 1$ in a structurally predictable way (temporal-derivative content). The residual open question is the structural *explanation*: why is D_W the wrong generator for the BW vacuum, signature-independently? We leave this as open question O4 in Section 10.

7.2 Pythagorean orthogonality and closed-form residual

Theorem 7.1 records the bit-exact signature-independence of the residual at $N_t = 1$. A post-L2-E PSLQ probe (`debug/h_local_residual_pslq_memo.md`, 2026-05-17) identified the structural mechanism behind that signature-independence: the residual decomposes as a Pythagorean sum, with one summand in the master Mellin engine M1 ring (Paper 32 § VIII; $\mathrm{Vol}(S^1)^2 = (2\pi)^2$ Hopf-base measure signature) and one summand purely rational.

Corollary 7.2 (Pythagorean orthogonality of H_{local} and D_L^W)**.** *At the Riemannian limit $N_t = 1$, the BW-aligned local Hamiltonian $H_{\mathrm{local}} = K_L^{\alpha,W}/\beta$ and the wedge-restricted Lorentzian Dirac D_L^W are Hilbert–Schmidt orthogonal,*

$$\langle H_{\mathrm{local}}, D_L^W \rangle_{\mathrm{HS}} \;:=\; \mathrm{Tr}\big(H_{\mathrm{local}}^\dagger\, D_L^W\big) \;=\; 0,$$

and the residual of Theorem 7.1 admits the closed form

$$\lVert H_{\mathrm{local}} - D_L^W \rVert_F^2 \;=\; \frac{\kappa_g^2 \cdot S(n_{\max})}{4\pi^2} \;+\; D(n_{\max}), \tag{30}$$

where κ_g is the witness surface-gravity normalisation, and

$$S(n_{\max}) = \frac{n_{\max}(n_{\max}+1)(n_{\max}+2)(2n_{\max}^2 + 4n_{\max} - 1)}{15},$$

$$D(n_{\max}) = \frac{n_{\max}(n_{\max}+1)(n_{\max}+2)(2n_{\max}+1)(2n_{\max}+3)}{20}$$

are pure rationals in $n_{\max}$.

The closed form is verified at $n_{\max} \in \{1, \ldots, 6\}$ by PSLQ at 100 dps with coefficient ceiling 10^6 on the basis $\{r^2, 1, 1/\pi^2\}$; every panel cell returns a clean integer relation. The rational coefficients S, D are recognisable as cumulative wedge-Casimir traces: $S(n_{\max})/4 = \sum_{\text{wedge}} m_j^2$ is the squared angular-momentum trace around the Hopf axis, and $D(n_{\max}) = \sum_{\text{wedge}}(n + \frac{1}{2})^2$ is the squared-Dirac-spectrum trace on the wedge.

Structural mechanism. $H_{\text{local}} = (\kappa_g/2\pi) \cdot \mathrm{diag}(\text{two\_m}_j > 0)$ is real and diagonal in the unfolded wedge spinor basis. The wedge-restricted Camporesi–Higuchi Dirac D_W^{GV} acts off-diagonally in the same basis: it intertwines the $\pm m_j$ chirality partners under the wedge basis change. The two operators live in orthogonal HS subspaces of $\mathcal{B}(K_W)$, hence the inner product vanishes. The diagonal/off-diagonal subspace decomposition is sketched at the basis level here and verified bit-exactly across the panel. The formal proof is a chirality-pairing cancellation: $D_W = \Pi_W \cdot |D_W|$ where $\Pi_W = \mathrm{diag}(\chi_i)$ is the wedge chirality grading with $\mathrm{Tr}\Pi_W = 0$ (equal-dimensional ± 1 eigenspaces), $|D_W|$ is Π_W-even, and H_{local} is Π_W-even. Hence $\mathrm{Tr}(H_{\text{local}}^\dagger D_W) = \mathrm{Tr}(\Pi_W \cdot M) = 0$ with $M = |D_W| H_{\text{local}}$ even under Π_W. Verified across 13 panel cells ($n_{\max} \in \{2, \ldots, 5\}$ Riemannian plus 9 Lorentzian $(n_{\max}, N_t)$ cells) with max residual 4×10^{-15} (numerical verification recorded in the project CHANGELOG, Sprint Pythagorean-orthogonality; the chirality-pairing proof above is self-contained).

Universality across six witnesses. The factorisation (30) is linear in κ_g^2: the diagonal piece $\|H_{\text{local}}\|_F^2$ scales as κ_g^2 via the global prefactor, and the off-diagonal piece $\|D_L^W\|_F^2$ is κ_g-independent (the Lorentzian Dirac construction of Section 2.2 contains no κ_g or β). The HS-orthogonality is therefore universal across the six modular witnesses of Section 6: all six share the BW-aligned $H_{\text{local}} = K_L^{\alpha,W}/\beta$ convention, differ only in $\kappa_g^{(w)}$, and produce bit-exact $\langle H_{\text{local}}^{(w)}, D_L^W \rangle_{\text{HS}} = 0$ at every cell of the 18-cell panel (six witnesses $\times$ $n_{\max} \in \{1, 2, 3\}$; max $|\langle H, D \rangle_{\text{HS}}| = 8.9 \times 10^{-16}$, machine precision). Source data: `debug/data/six_witness_hs_orthogonality.json`.

The $N_t > 1$ extension. The Pythagorean decomposition persists at $N_t > 1$ (verified bit-exactly across the panel $(n_{\max}, N_t) \in \{(1, 11), (1, 21), (2, 11), (3, 11), (3, 21), (4, 11)\}$). The temporal-derivative content $i\gamma^0 \otimes \partial_t$ in D_L enters the off-diagonal subspace, not the diagonal one, so the orthogonality is preserved. The $N_t > 1$ residual increment is a pure-rational shift driven by the finite-difference grid spacing; no new transcendentals are introduced. The master Mellin engine M1 ring is closed under the $N_t > 1$ extension.

Master Mellin engine reading. The residual sits entirely in the M1 sector of Paper 32 § VIII / Paper 18 § III.7's master Mellin engine: the sole transcendental is $1/\pi^2 = 1/\mathrm{Vol}(S^1)^2$, the squared Hopf-base-measure signature. This is structurally a $k = 0$ instantiation of the case-exhaustion theorem, where the orthogonal pair of cumulative Casimir traces (S in the K_α sector, D in the D_W sector) decompose into two non-overlapping subspaces of the wedge operator algebra. The residual does *not* join the framework's irreducible-but-natural list ($K = \pi(B + F - \Delta)$, Sprint MR-C L_2 next-order constant $c \approx 4.10932\ldots$, $S_{\text{full}}(\text{GS})$, Wolfenstein parameters); it is fully decomposable in M1, with explicit polynomial-in-$n_{\max}$ coefficients.

Formal proof (Sprint Pythagorean Orthogonality, 2026-05-23). The structural sketch above promotes to a formal theorem via a $\mathbb{Z}_2$ parity argument. Let Π_W denote the *wedge chirality grading* on $\mathcal{K}_W$ — the BBB Krein-chirality γ^5 of Paper 44 [32] § 3 restricted to the wedge. Then Π_W is a unitary self-adjoint involution ($\Pi_W^2 = I$, $\Pi_W^\dagger = \Pi_W$). Three structural inputs (Papers 42 [31] § 5, 7.3 BBB-axiom finding, and L2-F.1 unfolded-wedge basis change in `debug/h_local_residual_pslq_memo.md`):

(I1) *Even parity of $H_{\rm local}$:* $[\Pi_W, H_{\rm local}] = 0$. The geometric BW-$\alpha$ generator $K_\alpha^W = J_{\rm polar}^W$ acts on the wedge basis by integer eigenvalues $2m_j$ that depend only on the angular-momentum quantum number, not on the chirality.

(I2) *Odd parity of D_L^W:* $\{\Pi_W, D_L^W\} = 0$. Both factors of $D_L = i(\gamma^0 \otimes \partial_t + D_{\rm GV} \otimes I_{N_t})$ are Π_W-odd on the wedge: the BBB temporal gamma γ^0 anticommutes with γ^5 in the chiral basis (Paper 44 § 3), and the wedge-restricted Camporesi–Higuchi $D_W^{\rm GV}$ is Π_W-odd via the unfolded-wedge basis change (the diagonal-in-$(n_{\rm fock}, l, m_j)$ matrix elements of the truthful chirality-diagonal $D_{\rm GV}$ become off-diagonal cross-terms in the unfolded basis where Π_W swaps the two CH-chirality eigenvalues at fixed $(n_{\rm fock}, l, m_j)$).

Lemma 7.3 (Parity / HS-orthogonality)**.** *Let $\mathcal{H}$ be finite-dimensional, $\Pi : \mathcal{H} \to \mathcal{H}$ a unitary involution ($\Pi^2 = I$, $\Pi^\dagger = \Pi$), and $A, B \in \mathcal{B}(\mathcal{H})$ with $[\Pi, A] = 0$ and $\{\Pi, B\} = 0$. Then $\langle A, B\rangle_{\rm HS} := \mathrm{Tr}(A^\dagger B) = 0$.*

Proof. The trace is invariant under unitary conjugation: $\mathrm{Tr}(A^\dagger B) = \mathrm{Tr}(\Pi A^\dagger B \Pi^{-1}) = \mathrm{Tr}((\Pi A^\dagger \Pi^{-1})(\Pi B \Pi^{-1})) = \mathrm{Tr}(A^\dagger \cdot (-B)) = -\mathrm{Tr}(A^\dagger B)$, where we used $\Pi A^\dagger \Pi^{-1} = A^\dagger$ from $[\Pi, A] = 0$ together with $\Pi^\dagger = \Pi$, and $\Pi B \Pi^{-1} = -B$ from $\{\Pi, B\} = 0$. Hence $\mathrm{Tr}(A^\dagger B) = 0$. □

Theorem 7.4 (Pythagorean HS-orthogonality, formal closure)**.** *Under the construction of §2.2 at finite $(n_{\max}, N_t)$ and any modular witness in* $\{\mathrm{BW}, \mathrm{HH}, \mathrm{Sew}, \mathrm{Unruh}\}$,

$$\langle H_{\rm local}, D_L^W \rangle_{\rm HS} \;=\; \mathrm{Tr}_{\mathcal{K}_W}(H_{\rm local}^\dagger D_L^W) \;=\; 0$$

holds as an exact operator-algebraic identity. Corollary: the Pythagorean closed form $r^2 = \|H_{\rm local}\|_F^2 + \|D_L^W\|_F^2$ of (30) follows by expansion of $\|H_{\rm local} - D_L^W\|_F^2$.

Proof. Apply Lemma 7.3 with $\mathcal{H} = \mathcal{K}_W$, $\Pi = \Pi_W$, $A = H_{\rm local}$ (Input I1's even parity), $B = D_L^W$ (Input I2's odd parity). The lemma gives $\mathrm{Tr}(H_{\rm local}^\dagger D_L^W) = 0$ directly. □

The L2-F.1 numerical panel at 18 cells is empirical corroboration of the theorem at machine precision; the residuals at $\sim 10^{-14}$ are float64 roundoff in the matrix-element evaluation, not finite-cutoff bound noise. The *universality across six witnesses* is implied: the theorem statement is κ_g-independent (witnesses parameterise κ_g, which rescales $H_{\rm local}$ inside the Π_W-even subspace but cannot rotate it into the Π_W-odd subspace where D_L^W sits), so all six produce bit-exact zero by the same argument.

Honest scope (post-formal-proof). The theorem closes Paper 43's named follow-on (§ 10 O4) at finite $(n_{\max}, N_t)$ in the operator-algebraic sense. Three caveats remain: (i) the proof rests on the three structural inputs from Papers 42/43/44 (the Π_W identification with BBB γ^5 and the two parity statements I1, I2), each established in the cited papers but not re-derived here from first principles; (ii) the continuum limit $(n_{\max}, N_t, T) \to \infty$ requires Paper 47 [35]'s two-rate hybrid convergence at the norm-resolvent level, not the present operator-algebraic identity; (iii) the result lives on the wedge $\mathcal{K}_W$ at signature $(3,1)$, not on cross-manifold $\mathcal{T}_{S^3} \otimes \mathcal{T}_{\mathrm{Hardy}(S^5)}$ where Paper 24 § V's Coulomb/HO asymmetry blocks the construction.

7.3 The BBB universal-axiom finding on truthful $D_{\rm GV}$

We discuss here the structural reading of Proposition 2.6, the BBB universal-axiom failure $\{\chi, D\} \neq 0$ on the truthful Camporesi–Higuchi $D_{\rm GV}$.

7.3.1 The mutually-inconsistent triple

Three properties of the construction are mutually inconsistent in the sense that any two are compatible but the three together are not:

(P1) *Chirality-as-γ^5.* The GeoVac `FullDiracLabel.chirality` field is identified with the γ^5 eigenvalue in the chiral basis (Sprint L2-B convention, Section 2.1).

(P2) *Chirality-diagonal D_{GV}.* The Camporesi–Higuchi D_{GV} on $\mathcal{H}_{\mathrm{GV}}$ is diagonal with eigenvalue $\chi(n_{\mathrm{fock}} + 1/2)$, by construction (Paper 32 § III).

(P3) *BBB universal axiom $\{\chi, D\} = 0$.* Required of any indefinite spectral triple at any signature, per Bizi–Brouder–Besnard 2018 § 5(v).

The conjunction of (P1) and (P2) gives $[\gamma^5, D_{\mathrm{GV}}] = 0$ (both diagonal in the chirality index, hence commute), which contradicts (P3). The structural finding of Section 2.3 is that GeoVac's chirality-doubled spatial spinor bundle, equipped with the truthful Camporesi–Higuchi Dirac, fails (P3) under the chiral-basis convention adopted from Sprint L2-B.

7.3.2 Three resolutions and the chosen R1

Three resolutions are available; we follow Paper 42 R3.5 truthful-vs-offdiag trade in choosing R1.

(R1) *Accept the structural finding as documented scope.* Use truthful D_{GV}; the construction is a Krein-self-adjoint operator pair (Proposition 2.3) with the BBB-predicted-sign $(4, 6)$ relations on the J_L-axioms (Theorem 2.4), but does not constitute a full BBB indefinite spectral triple in the strict Sec 5(v) sense. Preserves the Riemannian-limit recovery bit-identically. This is the choice adopted by the present paper.

(R2) *Use the offdiag Camporesi–Higuchi Dirac.* The chirality-flipping off-diagonal variant $D_{\mathrm{GV}}^{\mathrm{offdiag}}$ from the present author's supporting module (`geovac.full_dirac_operator_system`) satisfies $\{\gamma^5, D_{\mathrm{GV}}^{\mathrm{offdiag}}\} = 0$ by construction, so the construction lifts to a full BBB Krein spectral triple. But $D_{\mathrm{GV}}^{\mathrm{offdiag}}$ breaks Riemannian-limit recovery bit-exactness (cf. Paper 42 R3.5 spinor-only-as-SDP-bound).

(R3) *Redefine γ^5 as the off-diagonal grading.* Build a different chirality operator on $\mathcal{H}_{\mathrm{GV}}$ that anticommutes with truthful D_{GV}. Breaks the chirality-as-γ^5 identification of Sprint L2-B; effectively rebuilds the basis-convention infrastructure from scratch.

7.3.3 Why R1 is consistent with the period closures

The wedge-modular construction of Sections 3 and 4 is $K_L^{\alpha,W}$*-driven and D_L-independent at the operator-action level.* The period closures $\sigma_{2\pi}^{L,\alpha}(O) = O$ and $\sigma_{2\pi}^{L,\mathrm{TT}}(a) = a$ depend on the integer spectrum of $K_L^{\alpha,W}$, not on D_L. The BBB universal-axiom failure on truthful D_{GV} therefore does not obstruct the period closures of Theorems 3.5 and 4.5, nor the flow conjugacy (3), nor the six-witness collapse.

7.3.4 Cross-track: Riemannian-side analog

Paper 32 § IV's Connes axiom audit at KO-dim 3 (Riemannian) verifies the relation $JD = +DJ$ exactly. There is no $\chi D = -D\chi$ axiom at KO-dim 3 because KO-dim 3 is odd and odd-dimensional spectral triples have no chirality grading. The BBB universal anticommutation $\{\chi, D\} = 0$ is therefore *introduced* at the $(3, 1)$ extension, and the structural finding is genuinely new at $(3, 1)$, not visible from the Riemannian side. The R1 resolution preserves both the Riemannian-limit bit-exactness and the period closures, at the cost of accepting that the (3, 1) extension under truthful D_{GV} + chirality-as-γ^5 is not a full BBB indefinite spectral triple in the strict Sec 5(v) sense.

7.4 Master Mellin engine M3 sub-mechanism: signature-independent

A Sprint L0 audit prediction in the present author's internal materials states:

> "M3 sub-mechanism of the master Mellin engine (vertex-parity Hurwitz / Dirichlet-L content in Camporesi–Higuchi vertex sums, per Paper 28 § QED-vertex) becomes structurally empty on chirality-symmetric spectrum at signature (3, 1)."

We tested this prediction on the chirality-symmetric truncation at $(3,1)$ in two natural conventions.

7.4.1 Reading A: $n_{\mathbf{fock}}$-parity (Paper 28 convention)

The Paper 28 § QED-vertex vertex-parity sum reads parity from the Fock principal quantum number n_{fock}:

$$D^N_{\mathrm{even}}(4) - D^N_{\mathrm{odd}}(4) \;=\; \sum_{n_{\mathrm{fock}}=1}^{N} \frac{g_n \cdot (-1)^{n_{\mathrm{fock}}}}{(n_{\mathrm{fock}} + 1/2)^4},$$

with $g_n = 2n(n+1)$ the full-Dirac degeneracy per level (summed across both chiralities). This is the convention that accesses Catalan's constant G and Dirichlet $\beta(4)$ via quarter-integer Hurwitz at $a = 3/4$ and $a = 5/4$.

On the chirality-symmetric truncation at $(3,1)$, each n_{fock} level contributes its full g_n — both chirality blocks together. **Therefore $D_{\mathrm{even}} - D_{\mathrm{odd}}$ at $(3,1)$ is identical to the Riemannian value.** Bit-exact at $n_{\max} \in \{1,2,3,4,5\}$.

7.4.2 Reading B: chirality-pairing

Pairing "even" with $\chi = +1$ and "odd" with $\chi = -1$ gives $D_+ - D_- = 0$ identically on any chirality-symmetric truncation, by construction.

7.4.3 Verdict

The L0 prediction is **convention-dependent**. Under Reading A, the prediction is falsified: M3 does *not* trivialise at $(3,1)$ on the chirality-symmetric truncation. Under Reading B, the prediction holds tautologically by chirality symmetry alone, but Reading B does not access Paper 28's M3 content (Catalan G, $\beta(4)$).

The cleaner structural reading is: *The master Mellin engine M3 sub-mechanism is sectional in n_{fock}-parity, not in spacetime signature.* The BBB sign-flip $\{J, \chi\} = 0$ at $(3,1)$ is a spectral-triple axiom statement, not a vertex-parity-sum statement, and does not directly imply M3 trivialisation on chirality-symmetric truncations because Paper 28's vertex-parity sum uses a chirality-independent parity label.

This refines the master Mellin engine domain partition of Paper 32 § VIII without disturbing the case-exhaustion theorem: the sub-mechanism classification at the level of which transcendental appears (M1 $= 2\pi\cdot\mathbb{Q}$, M2 $= \sqrt{\pi}\,\mathbb{Q}$, M3 = Hurwitz / Catalan / Dirichlet-L) is unchanged; the refinement is that the M3 mechanism's natural domain is the n_{fock}-parity sectional structure, not the spacetime signature.

8 Riemannian-limit recovery: this is a lift, not a parallel construction

Theorem 8.1 (Riemannian-limit recovery at $N_t = 1$)**.** *At $N_t = 1$, the Sprint L2-E Lorentzian constructions of the present paper reduce* bit-identically *(Frobenius residual exactly* 0.0*, not merely machine precision) to the Paper 42 Riemannian-side constructions: $P_{W_L}|_{N_t=1} = P_W^{\text{spatial}}$; $K_L^\alpha|_{N_t=1} = K_\alpha^{\text{spatial}}$; $K_L^{\alpha,W}|_{N_t=1} = K_\alpha^W$; $\rho_W^L|_{N_t=1} = \rho_W$; $\Delta_L|_{N_t=1} = \Delta$; $K_L^{\text{TT}}|_{N_t=1} = K_{\text{TT}}$.*

Proof. The reduction is by construction: at $N_t = 1$, the Kronecker products $\cdot \otimes I_{N_t}$ reduce to $\cdot \otimes I_1 = \cdot$, and the temporal projector $P_{t\geq 0}|_{N_t=1} = I_1$. Every Krein-side construction collapses to its spatial-side analog in Paper 42. Bit-identicality (not merely machine precision) follows from the integer-arithmetic structure of the Kronecker-with-I_1 reduction. □

Significance. The Riemannian-limit recovery is the load-bearing falsifier that ensures the present paper is the *Lorentzian extension* of Paper 42, not a parallel-but-different construction. If the reduction failed (i.e., if $K_L^\alpha|_{N_t=1} \neq K_\alpha^{\text{spatial}}$), it would indicate the Lorentzian-side wedge and modular structures are categorically distinct from the Riemannian-side ones even at the Riemannian limit — and that would re-open Paper 42's closure at the operator-system level.

The bit-identical reduction shows the Lorentzian extension is *structurally faithful* to Paper 42 by construction. The temporal slot adds additional content at $N_t > 1$ (the temporal-derivative piece of D_L, the larger wedge sub-Hilbert space, etc.) that is invisible at $N_t = 1$; at $N_t > 1$, the Riemannian-side Paper 42 result is enriched by Lorentzian content without overwriting it.

Paper 42 not superseded. The present paper does not supersede Paper 42: Paper 42 is the Riemannian closure of the four-witness Wick-rotation theorem on the SU(2) truncated triple, and Paper 43 is the Lorentzian extension on the Krein-space truncated triple. Together they constitute the four-witness Wick-rotation arc at the operator-system level (finite cutoff), Paper 42 the Riemannian half and Paper 43 the Lorentzian half.

9 Honest scope

9.1 Finite cutoff only

The main theorem (Theorem 1.1) and the related period-closure theorems (Theorems 3.5, 4.5, 5.1) all hold at *finite* $n_{\max}$ and finite N_t. They are not analytic-limit statements. The closure is a finite-cutoff bit-exact algebraic identity, with the structural reason being the integer spectrum of $K_L^{\alpha,W}$ on the spinor harmonic basis. No Gromov–Hausdorff limit is invoked or required; the period-closure identities hold at every cutoff verified.

9.2 Lorentzian-propinquity continuum closure remains open

A Lorentzian-side analog of Paper 38 [29]'s qualitative-rate state-space GH convergence — i.e., the statement that the truncated Krein-space spectral triples $\mathcal{T}^L_{n_{\max},N_t}$ converge in some quantum analog of the Latrémolière propinquity to a continuum Lorentzian spectral triple on $S^3 \times \mathbb{R}$ — remains open.

No Lorentzian-side analog of the Latrémolière quantum Gromov–Hausdorff propinquity [14, 15] exists in the published literature as of May 2026. The existing propinquity-convergence literature (Latrémolière [14, 15], Leimbach–van Suijlekom [16] for the torus, Hekkelman–McDonald [13], Toyota [24], Farsi–Latrémolière [8, 9]) all operates on Riemannian spaces. The twist-morphism construction of Nieuviarts [18–20] is restricted to even-dimensional manifolds (Definition 2.2 of [19]) and does not apply to the $S^3 = \mathrm{SU}(2)$ case (KO-dim 3, odd).

A Lorentzian propinquity construction at signature $(3,1)$ on $S^3 \times \mathbb{R}$ is an original NCG-mathematics work at multi-month scale; it is the named Sprint L3 follow-up of the present author's framework documentation and is not addressed by the present paper.

Update post-Sprint L3b / L3c (May 2026). After this paper was drafted, the Lorentzian-propinquity program was investigated in two stages and *descoped*: Paper 45 [33] proves a *degeneracy theorem* — the natural Lorentzian-propinquity seminorm on the truncated $\mathrm{SU}(2) \times U(1)_T$ Krein spectral triple vanishes identically, so the natural Lorentzian-propinquity route is closed, not established — and Paper 46 [34]'s strong-form construction is correspondingly descoped, the natural-substrate seminorm being degenerate (an enlarged-substrate repair is stated as a target, not a theorem). The Lorentzian *quantum-metric / propinquity* convergence is therefore an *open* research target, not a completed result. What *does* survive is signature-agnostic: Paper 45's product-carrier rate on $S^3 \times S^1_T$ at the canonical compact temporal radius $T = 2\pi$ (a Riemannian GH-convergence statement, not a Lorentzian one), and Paper 47 [35]'s *norm-resolvent* de-compactification arrow $T \to \infty$ to the Lorentzian Dirac on $S^3 \times \mathbb{R}_t$ with exponential tail $O(e^{-|\mathrm{Im}\, z|T/2})$. The three-carrier identification of Paper 47 § 6 identifies the present paper's *bounded Dirichlet* carrier $S^3 \times [-T_{\max}, T_{\max}]$ and the *periodic compact* carrier $S^3 \times S^1_T$ as two compact approximation schemes converging in the norm-resolvent sense to the same non-compact Lorentzian Dirac on $S^3 \times \mathbb{R}_t$; that spectral / operator-algebraic identification stands. The *propinquity*-level Lorentzian closure does not.

The *propinquity-level* continuum closure on a non-compact carrier (a single uniform Latrémolière bound to $\mathcal{T}^L_{S^3\times\mathbb{R}_t}$, rather than the two-rate hybrid) *remains open*, requiring a non-compact extension of Latrémolière propinquity that is not in the published literature as of May 2026 — structurally, because the finite-cutoff modular flow is compact and hence carries only a Euclidean metric, so the Lorentzian signature can appear only in the non-compact limit (Paper 45 [33] § open Q1; see O1 below). This is named G1-genuine on the non-compact substrate (or equivalently G2-metric) in Paper 45 § 1.4 / Paper 47 § 8.

9.3 Cross-manifold W2b not addressed

The cross-manifold tensor product $\mathcal{T}_{S^3} \otimes \mathcal{T}_{\mathrm{Hardy}(S^5)}$ of the round-S^3 Camporesi–Higuchi triple with the Bargmann–Segal Hardy-sector spectral triple on S^5 remains structurally open ("W2b-medium" in the present author's multi-focal-composition wall taxonomy). The blocker is at the NCG-framework level: no published construction handles a Riemannian-times-Hardy-sector tensor product spectral triple in a uniform propinquity sense, and the Coulomb/HO category mismatch of Paper 24 § V [27] re-emerges at the spectral-triple level. This is a distinct frontier from the present $(3,1)$ extension within $\mathcal{T}_{S^3}$.

9.4 Calibration data (W3) not addressed

The present paper is structurally additive at the spectral-triple-machinery level: it extends Paper 42 by lifting the operator-system Wick-rotation theorem to signature $(3,1)$. The *inner-factor input-data* question (Yukawa Dirichlet ring, calibration constants; "W3 / second packing axiom" in the present author's internal materials) is untouched by the present construction. The Lorentzian extension does not generate calibration data; it extends the structural skeleton.

9.5 Bit-exactness reflects construction, not numerics

The bit-exact residuals reported in Tables 1, 2, 3, and elsewhere are not numerical accident. They reflect the structural fact that the construction reduces to operations on real integer permutation matrices (J_{spatial}, U_L, R_{polar}), simple rationals ($\chi(n+1/2)$ on D_{GV}; $\pm(2/\Delta_t)$ on ∂_t at dyadic $T_{\max}$), the global i factor, and Kronecker products with I_{N_t}. None of these operations introduces transcendental content or accumulates floating-point error, so the period-closure identities — which depend on integer spectrum of $K_L^{\alpha,W}$ and the integer-arithmetic structure of $J = J_{\text{spatial}} \otimes I_{N_t}$ — hold to within sub-machine precision at every panel cell.

10 Open questions

O1. Lorentzian propinquity continuum closure (Sprint L3). The Lorentzian-side analog of Paper 38 [29]'s qualitative-rate state-space GH convergence remains open *at the continuum / non-compact level.* The *finite-cutoff* side is now understood structurally (Paper 45 [33] § open Q1, 2026-06): the truncated Bisognano–Wichmann boost is *compact* (the modular generator $K = \text{diag}(2m_j)$ has integer spectrum, so $e^{2\pi i K} = I$ at every finite cutoff and the modular flow is the 2π-periodic KMS circle, not a non-compact boost). A compact circle carries only a Euclidean (forward-triangle) metric, so at every finite cutoff the quantum metric is Euclidean and the Lorentzian signature is the Wick rotation $t \to it$ of that circle — a convention, not a finite-cutoff metric. This is precisely *why* the open continuum closure *requires a non-compact extension* (below): the reverse-triangle Lorentzian signature lives only in the non-compact ($n_{\max} \to \infty$, type-III) limit. No Lorentzian propinquity construction is published in the spectral-truncation literature as of May 2026; the continuum construction is original NCG-mathematics work at multi-month scale, the natural follow-up to the finite-cutoff closure of the present paper.

O2. Cross-manifold $\mathcal{T}_{S^3} \otimes \mathcal{T}_{\text{Hardy}(S^5)}$. The cross-manifold tensor product of the Riemannian Camporesi–Higuchi triple with the Bargmann–Segal holomorphic-sector spectral triple on S^5 remains structurally open, blocked at the NCG-framework level by the Coulomb/HO category mismatch of Paper 24 § V [27].

O3. Inner-factor calibration (W3, second packing axiom). The inner-factor input-data question (Yukawa Dirichlet ring, calibration constants) is structurally distinct from the present spectral-triple-machinery extension. The present paper does not address W3.

O4. Stronger H_{local} characterization: why is $H_{\text{local}} \neq D_W$ signature-independent? Section 7.1 sharpens Paper 42's open question O3 from "open about signature-dependence" to "signature-independent at the Riemannian limit, refined at $N_t > 1$ by temporal-derivative content". Section 7.2 (Corollary 7.2) further sharpens the residual to a structural Pythagorean identity: H_{local} and D_L^W live in mutually-orthogonal Hilbert–Schmidt subspaces of $\mathcal{B}(\mathcal{H}_W)$, with the squared residual decomposing in the master Mellin engine M1 ring. The diagonal/off-diagonal subspace decomposition is now **formally closed** by Theorem 7.4 (added 2026-05-23 via the Sprint Pythagorean Orthogonality formal-proof closure documented in `debug/sprint_pythagorean_orthogonal` a $\mathbb{Z}_2$ wedge-chirality grading Π_W exists with H_{local} even and D_L^W odd, and the HS-orthogonality follows directly from the cyclicity-of-trace / unitary-conjugation argument (Lemma 7.3). The bit-exact L2-F.1 numerical panel at 18 cells is empirical corroboration of the now-algebraic identity, not finite-cutoff bound noise. The deeper structural-explanation question — why is the spectral-action Dirac not the modular generator on the Bisognano–Wichmann vacuum at either signature, and why does it fail by sitting in a literally-orthogonal HS subspace? — remains open. The Connes–Rovelli [6] thermal-time hypothesis and the Chamseddine–Connes [4] spectral-action paradigm point to two distinct natural generators on the wedge KMS state; the Pythagorean orthogonality observed here suggests the distinction is structural and basis-canonical, not a small perturbative effect. The S^3-specificity of this construction (Paper 24 § 5.3 [27] fourth-layer Coulomb/HO asymmetry) places it in the same family as the spectrum-computing L_0, the calibration π, and the Wilson matter coupling — each of which has no clean Bargmann–Segal S^5 analog at the present state of the framework.

O5. Non-tracial wedge states (Paper 42 O2 lifted). The wedge KMS state ρ_W^L of the present paper is a tracial-Gibbs state on the wedge sub-Hilbert space. Non-tracial wedge states (KMS states with respect to a non-tracial reference) would require additional infrastructure (e.g., Tomita's Type III factor framework). The extension is a Paper 42 open question O2 lifted to $(3,1)$; the present paper does not address it.

O6. Higher-rank compact non-abelian extensions (Paper 42 O4 lifted). Paper 40 [30] establishes a universal $4/\pi$ rate constant for state-space Gromov–Hausdorff convergence on all compact connected Lie groups with bi-invariant metric. The modular structure of the present paper plausibly extends to higher-rank cases; the integer spectrum of the BW-α generator $K_L^{\alpha,W}$ is a property of SU(2) specifically (where the half-integer weights double to integers), and the higher-rank Lorentzian analog requires verifying that the relevant Cartan generator along the wedge axis still has the right spectral property at signature $(3,1)$.

O7. BBB-universal-axiom-compatible construction. The Sprint L2-D R3 resolution (Section 7.3) — redefine γ^5 as an off-diagonal grading on $\mathcal{H}_{\mathrm{GV}}$ that anticommutes with truthful D_{GV} — remains a possible direction for a "full BBB Krein spectral triple in the strict Sec 5(v) sense" with truthful D_{GV}. The Sprint L2-D R2 resolution — use offdiag D_{GV}, which satisfies $\{\gamma^5, D_{\mathrm{GV}}^{\mathrm{offdiag}}\} = 0$ but breaks Riemannian-limit recovery bit-exactness — is similarly available. Both are distinct directions from the R1 truthful-D_{GV} choice of the present paper.

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